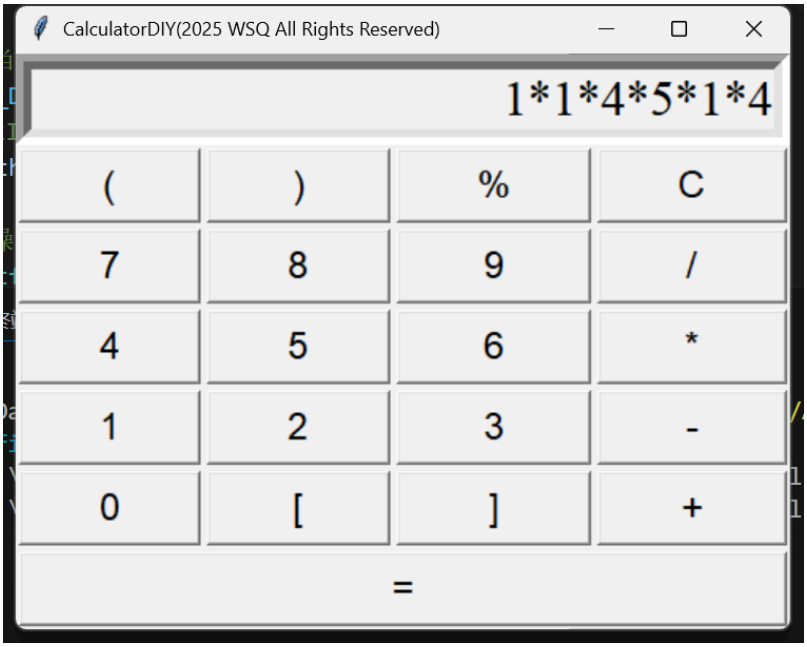
Lab 2 Postfix Calculator

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Part I. Lab Structure

- Head files
  - `postfix_cal.h`
    Calculating the postfix expression.
  - `in2post.h`
    Transform the infix expression to the postfix expression.
  - `calculator.h`
    Implementation of the Calculator, we first transform the expression string from postfix expression, then use `postfix_cal.h` to calculate the result.
  - `string2vec.h`
- Test `cpp` files
  - `postfix_cal_test.cpp`
    Test for `postfix_cal.h`
  - `in2post_test.cpp`
    Test for `in2post.h`
  - `caltest.cpp`
    Test for `calculator.h`
- `calculator_core.cpp`
  generate `calculator.dll`
- `Makefile`
- `calculator_gui.py`
  Python GUI for the Calculator

Part II.Quick Start

Testing

```
mingw32-make
```

```
.\postfix_cal_test
```

```
Test 1:
6 4 + 7 2 - /
result = 2
-----------Test 1 passed------------
Test 2:
6 7 * 5 8 + %
result = 3
-----------Test 2 passed------------
Test 3:
1 3 + 2 4 + * 6 - 8 2 - /
result = 3
-----------Test 3 passed------------
```

```
.\in2post_test
```

```
Test 1:
Infix: ( 6 + 2 ) / 4
Postfix: 6 2 + 4 /
--------------------------------------
Test 2:
Infix: [ ( 1 + 3 ) * ( 2 + 4 ) - 6 ]
Postfix: 1 3 + 2 4 + * 6 -
--------------------------------------
Test 3:
Infix: ( 1 + 9 ) * ( 8 + 9 ) / ( 6 - 4 )
Postfix: 1 9 + 8 9 + * 6 4 - /
--------------------------------------
Test 4:
Infix: [ ( 1 + 3 ) * ( 2 + 4 ) - 6 ] / ( 9 - 7 ) % 2
Postfix: 1 3 + 2 4 + * 6 - 9 7 - / 2 %
--------------------------------------
Test 5:
Infix: 5 - 2 + 1
Postfix: 5 2 - 1 +
--------------------------------------
```

```
.\caltest
```

```
Test 1:
4+8*2 = 20
True
--------------------------------------
Test 2:
1 * 1 * 4 * 5 * 1 * 4 = 80
True
--------------------------------------
Test 3:
[(4+3-1)/(7%5)-4]*10/(2-5+8) = -2
True
--------------------------------------
```

Calculator GUI

```
python calculator_gui.py
```

> **Vibe Coding:**
> I use the VSCode Copilot to help write the GUI python file